International Journal of Cardiology 258 (2018) 59–67

Contents lists available at ScienceDirect

International Journal of Cardiology

journal homepage: www.elsevier.com/locate/ijcard

Perioperative aspirin therapy in non-cardiac surgery: A systematic review and meta-analysis of randomized controlled trials☆

Georg Wolff [a,d,*], Eliano Pio Navarese [d,e], Maximilian Brockmeyer [a,d], Yingfeng Lin [a,d], Athanasios Karathanos [a], Michalina Kołodziejczak [b,d], Jacek Kubica [b,d], Amin Polzin [a], Tobias Zeus [a], Ralf Westenfeld [a], Felicita Andreotti [c], Malte Kelm [a,d], Volker Schulze [a,d]

[a] Department of Internal Medicine, Division of Cardiology, Pulmonology and Vascular Medicine, Heinrich-Heine-University, Düsseldorf, Germany
[b] Department of Cardiology and Internal Medicine, Nicolaus Copernicus University, Bydgoszcz, Poland
[c] Department of Cardiovascular Science, Catholic University of the Sacred Heart, Rome, Italy
[d] Systematic Investigation and Research on Interventions and Outcomes (SIRIO)-Medicine Research Network
[e] Interventional Cardiology and Cardiovascular Medicine Research, Inova Heart And Vascular Institute, Fairfax Medical Campus, Falls Church, VA, USA

ARTICLE INFO

Article history:
Received 13 August 2017
Received in revised form 9 December 2017
Accepted 21 December 2017

Keywords:
Aspirin
Surgery
Bleeding
Mortality
Myocardial infarction
Cardiovascular disease
Thromboembolism
Stroke

ABSTRACT

Background: Aspirin is a key element in prevention of cardiovascular and thromboembolic events. During non-cardiac surgery however, its balance of bleeding risks and benefits remains unclear.
Methods: A systematic review and meta-analysis of randomized controlled trials was performed. Online databases were screened for clinical trials randomizing aspirin to no aspirin therapy in non-cardiac surgery. Clinical outcomes of all-cause mortality and cardiovascular mortality, arterial ischemic events, venous thromboembolic events and bleeding events were separately evaluated.
Results: Seven RCTs comprising 28,302 patients were included. All-cause mortality (3.7% vs. 3.8%; odds ratio (OR) 0.97, CI 0.86–1.10) and cardiovascular mortality (2.0% vs. 2.1%, OR 0.92; CI 0.78–1.09) were not different in aspirin vs. no aspirin groups. Arterial ischemic events showed no differences, including myocardial infarction (2.5% (aspirin) vs. 2.5% (no aspirin)), cerebrovascular events (0.6% (aspirin) vs. 0.6% (no aspirin)) and peripheral arterial events (0.2% (aspirin) vs. 0.3% (no aspirin)). Aspirin significantly reduced the risk for venous thromboembolic events (VTE; 1.5% (aspirin) vs. 2.0% (no aspirin); OR 0.74, CI 0.59–0.94, $p = 0.02$). Perioperative major bleeding was significantly more frequent in aspirin groups (4.4% vs. 3.7%; OR 1.18, CI 1.05 to 1.33, $p = 0.007$).
Conclusion: Aspirin remained neutral with respect to overall survival, cardiovascular mortality and arterial ischemic events. It reduced venous thromboembolic events at the expense of perioperative major bleedings. Thus, this analysis supports recommendations against perioperative aspirin continuation/initiation in cardiovascular disease patients at intermediate risk, as well as recommendations of aspirin for VTE prophylaxis in orthopedic patients only.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

An estimated 4% of the world population has a surgical procedure performed every year [1,2] — with a rising tendency [3]. Surgical procedures carry an inherent risk for complications, which depend on the type and quality of procedure and anesthesia, but also on patient comorbidities and safety precautions. Overall complication rate estimates range from 3 to 11% [4,5] to up to 40% [6], and among the leading causes – apart from technique-related complications – are cardiovascular adverse events and bleeding events [5].

Aspirin therapy for chronic platelet inhibition represents a cornerstone in the prevention of cardiovascular disease (CVD) related events. Irreversible cyclooxygenase-I inhibition reduces platelet activation by thromboxane A_2 [7,8] and thus aggregation in arterial and venous

Abbreviations: ACC, American College of Cardiology; ACCP, American College of Chest Physicians; AHA, American Heart Association; ASA, acetyl salicylic acid; BARC, Bleeding Academic Research Consortium; CABG, coronary artery bypass graft; CI, confidence interval; CVD, cardiovascular disease; DM, diabetes mellitus; ESC, European Society of Cardiology; HLP, hyperlipoproteinemia; HTN, hypertension; LOE, level of evidence; M–H, Mantel–Haenszel; MI, myocardial infarction; n/a, not available; NSAR, non-steroid anti-rheumatic drug; OR, odds ratio; PAOD, peripheral artery occlusive disease; PCI, percutaneous coronary intervention; PRISMA, Preferred Reporting Items for Systematic reviews and Meta-Analyses; RCT, randomized controlled trial; TIA, transient ischemic attack; VTE, venous thromboembolism.

☆ All authors take responsibility for all aspects of the reliability and freedom from bias of the data presented and their discussed interpretation.

* Corresponding author at: Department of Internal Medicine, Division of Cardiology, Pulmonology and Vascular Medicine, Heinrich-Heine-University, Moorenstr. 5, 40225 Düsseldorf, Germany.
E-mail address: georg.wolff@med.uni-duesseldorf.de (G. Wolff).

https://doi.org/10.1016/j.ijcard.2017.12.088
0167-5273/© 2017 Elsevier B.V. All rights reserved.

vessels. It more effectively reduces arterial ischemic events [9–14] in secondary prevention of CVD than it increases bleeding risk [10,11,15], making it a guideline-recommended standard for CVD patients [12, 13]. In the setting of non-cardiac surgery however – where bleeding represents a serious risk factor – aspirin's perioperative CVD prevention and prophylaxis of thromboembolism [16–19] must be weighed against its bleeding risks.

This delicate risk/benefit balance in the perioperative setting is not clear. We here aimed to comprehensively analyze and differentiate aspirin effects in non-cardiac surgical patients in a review and meta-analysis of randomized controlled trials (RCTs).

2. Materials and methods

This meta-analysis was performed according to established methods recommended by the Cochrane Collaboration and in compliance with the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) statement for reporting systematic reviews and meta-analyses in health care interventions [20,21]. All screening, study selection, data extraction and analysis processes were performed by independent investigators, who were not personally involved in any of the included trials or had any other conflict of interest on the topic.

2.1. Study design and endpoint selection

This analysis was designed to investigate cardiovascular and thromboembolic benefits and bleeding risks of perioperative aspirin therapy during non-cardiac surgery. All prospective randomized controlled clinical trials of aspirin vs. no aspirin therapy in non-cardiac surgery were included, which featured assessment of the primary outcome of all-cause mortality and were published in English language and in full text. Secondary outcomes were cardiovascular mortality, myocardial infarction, cerebrovascular events (as a combination of ischemic stroke and transient ischemic attack (TIA)), thromboembolic events (including venous thrombotic events and pulmonary embolism), peripheral arterial events and major bleeding.

2.2. Data sources, search strategy and study selection

Medline, PubMed, Web of Science, Cochrane Central Register of Controlled Trials (CENTRAL), Google Scholar, and Embase databases, along with the websites www.medscape.com/cardiology, www.clinicaltrials.gov, www.clinicaltrialresults.org and www.cardiosource.org, were systematically searched (GW, MB, YL, and MiK) up until July 2017 for relevant published trial reports. Search keywords included combinations of: aspirin, acetyl salicylic acid, ASA, randomized controlled trial, surgery, operation, bleeding. A bibliography search within landmark articles, meta-analyses and guidelines of medical societies on the subject was additionally performed and relevant trials were added. All articles were primarily screened at the title/abstract level and then retrieved as full text reports. Studies positively evaluated during eligibility assessment were finally selected for inclusion (GW, MB, YL, MiK).

2.3. Data collection and quality assessment

Data from included trials were abstracted into prespecified forms (MB, YL, GW, VS) and analyzed according to the intention-to-treat principle, where possible. Internal validity was ensured by cross-checking between investigators; divergences were resolved by consensus after discussion in the group (MB, YL, GW, EPN, VS). Bias assessment was performed by two unblinded investigators (MB, YL) according to the Cochrane Collaboration guidelines [21] and was again cross-checked for errors.

2.4. Secondary outcome definitions

Secondary outcome definitions showed heterogeneity across studies (Supplementary Table 3). Endpoints were analyzed as reported. For bleeding outcome, an event definition of *major bleeding* was used where available (corresponding to BARC >2 (Bleeding Academic Research Consortium, [22])), or was assumed (e.g. life-threatening, requiring transfusion, requiring reoperation etc.).

2.5. Data synthesis and statistical analyses

Odds ratios (ORs) and 95% confidence intervals (CIs) were used as summary statistics. Heterogeneity was assessed by the Cochran's Q test [23]. Statistical heterogeneity was summarized by the I^2 statistic, which quantifies the percentage of variation in study results that is due to heterogeneity rather than to chance [24]. I^2 values >20% were judged to indicate substantial heterogeneity, which prompted use of the DerSimonian and Laird random-effects model [25], instead of the less conservative fixed-effects model. Sensitivity analyses were performed to account for trial heterogeneity and further ascertain validity of the pooled analyses. The statistical level of significance of the Cochran–Mantel–Haenszel statistics estimate for the summary treatment effect was assumed at a 2-tailed p-value of <0.05. Review Manager version 5.3 (The Nordic Cochrane Centre, Copenhagen, Denmark), Microsoft Excel and SPSS version 23 (SPSS, Chicago, Illinois) were used for statistical computations.

3. Results

3.1. Study selection, trial protocols and patient populations

The process of article screening and selection is described in a PRISMA flow chart (Supplementary Fig. 1). Primary searches revealed a total of 9475 sources/reports, based on title and abstract, any duplicates and non-clinical studies were removed, and 72 studies were then subsequently evaluated and condensed to seven relevant prospective randomized controlled trials [26–32]. These were published between 1993 and 2014 and enrolled a total of 28,302 patients.

Study characteristics and randomization details are reported in Table 1: the largest included trials were the Pulmonary Embolism Prevention (PEP) trial [32] with 17,444 and the PeriOperative ISchemic Evaluation 2 trial (POISE-2) [27] with 10,010 patients; the smallest trials were APAP and Nielsen et al. with just over 50 patients [26,30]. Postoperative follow-up was one month in the majority of trials.

A wide range of procedures mainly corresponding to intermediate-cardiovascular-risk surgery [1] were performed (Table 1) in the trials: abdominal surgery (cholecystectomy, gastrectomy, hernia repair, bowel and colorectal surgery, oncologic surgery), orthopedic surgery (knee and hip surgery), urologic/prostate surgery, carotid endarterectomy, and retroperitoneal surgery (nephrectomy).

Patient characteristics of the included studies are shown in Supplementary Table 1. 10,858 patients in APAP, Lindblad et al., Nielsen et al., Oscarsson et al., STRATAGEM and POISE-2 [26–31] were assumed to suffer from CVD at average intermediate cardiovascular risk, as they were either on long-term aspirin therapy for prevention of CVD (Table 1) or were directly characterized as such in the published report. PEP patients were considered to be at low cardiovascular risk, as they were mostly lacking pre-randomization aspirin therapy [32]. Aspirin dosage varied from 75 mg to 300 mg daily. Six trials prescribed it for CVD prevention [26–31], PEP used it for prevention of venous thromboembolism [32]. Six studies randomized aspirin to placebo [27–32], APAP to discontinuation of aspirin [26].

Risk of bias of all included studies is depicted in Supplementary Table 2: trial quality in general was high, four trials used a multicenter design.

Details on secondary endpoint definitions and reporting of all included studies are listed in Supplementary Table 3.

3.2. All-cause mortality and cardiovascular mortality

The primary outcome of all-cause mortality was reported in all trials and all 28,302 patients [26–32]. The pooled meta-analysis showed similar rates of all-cause mortality (3.7% (aspirin) vs. 3.8% (no aspirin); OR 0.97 with CI 0.86–1.10; $I^2 = 0\%$; $p = 0.61$; Fig. 1A). Sensitivity analyses were performed to account for patient population differences (exclusion of the low-risk CVD population in PEP, Supplementary Fig. 2A) and heterogeneity in trial sizes (exclusion of heavy weight trials PEP and POISE-2, Supplementary Fig. 2B), however no significant differences in survival were found in either analysis.

All seven trials with 28,302 patients reported cardiovascular mortality [26–32] (Fig. 1B). The pooled analysis showed no difference for cardiovascular mortality (2.0% (aspirin) vs. 2.1% (no aspirin); OR 0.92 with CI 0.78–1.09; $I^2 = 0\%$, $p = 0.33$). When excluding the PEP trial for its low-CVD-risk population, results remained similar (sensitivity analysis, Supplementary Fig. 3A).

Table 1
Characteristics of included studies.

Study	Journal	Year	Patient count	Antiplatelet therapy before randomization	Randomization	ASA study indication	Type of surgery	Additional anticoagulants	Follow-up
APAP [26,59]	Langenbeck's Arch. Surg.	2012	52	Long-term ASA for prevention of CVD	Continued ASA 100 mg vs. no ASA (from 5 days before until 5 days after surgery)	Prevention of CVD	Cholecystectomy (25%), inguinal hernia repair (44%), colonic/colorectal surgery (31%)	n/a	3 months
Lindblad et al. [28]	Stroke	1993	232	None	ASA 75 mg vs. placebo (from evening before surgery for 6 months)	Prevention of vascular neurologic complications	Carotid endarterectomy	n/a	1 month
Nielsen et al. [30]	Scand. J. Urol. Nephrol.	2000	53	Long-term ASA for prevention of CVD	Continued ASA 150–300 mg vs. placebo (from 10 days before surgery until postoperative catheter removal)	Prevention of CVD	Transurethral prostatectomy	0%	30 days
Oscarsson et al. [31]	Br. J. Anaesth.	2010	220	Long-term ASA for prevention of CVD	ASA 75 mg vs. placebo (from 7 days before until 3 days after surgery)	Prevention of CVD	Bowel surgery, gastric surgery, prostate surgery (open or transurethral), cystectomy, nephrectomy, hip or knee arthroplasty, and intra-abdominal or pelvic cancer surgery	n/a	30 days
PEP [32]	Lancet	2000	17,444	9% ASA/NSAR within 48 h before randomization	ASA 160 mg vs. Placebo (started preoperatively until 35 days after surgery)	Prevention of venous thromboembolism	Hip-fracture, arthroplasty of knee or hip	44% (not specified)	35 days
POISE II [27]	N. Engl. J. Med.	2014	10,010	- 44% on long-term ASA for prevention of CVD, paused 3 days before randomization (continuation stratum) - 56% naïve (initiation stratum)	ASA 100 mg vs. placebo (started preoperatively until 7 days/30 days after surgery in continuation/initiation stratum)	Prevention of CVD	Orthopedic (39%), general (27%), urologic/gynecologic (17%), vascular (6%), thoracic (6%) and other	65% (not specified)	30 days
STRATAGEM [29]	Br. J. Anaesth.	2011	291	Long-term anti-platelet agents (72% ASA, 32% clopidogrel) for prevention of CVD	ASA 75 mg vs. placebo (from 10 days before until surgery)	Prevention of CVD	Orthopedic (52%), abdominal (21%), urologic (15%) and other	100% LMWH	30 days

Characteristics of all included randomized controlled trials. ASA = aspirin, acetylsalicylic acid; CVD = cardiovascular disease; n/a = not available; NSAR = non-steroid anti-rheumatic drug.

3.3. Arterial ischemic events

3.3.1. Myocardial infarction

All included trials assessed perioperative myocardial infarction [26–32]. Cumulative analysis resulted in similar OR for myocardial infarction in aspirin vs. no aspirin groups, albeit with considerable heterogeneity due to differing results in PEP and POISE-2 (2.5% (aspirin) vs. 2.5% (no aspirin); OR 1.00 with CI 0.66–1.53; I^2 = 38%, p = 0.99; Fig. 1C). When excluding the PEP trial for its low-risk CVD population, results remained similar and heterogeneity was reduced (I^2 = 5%, Supplementary Fig. 3B).

3.3.2. Cerebrovascular events: transient ischemic attack and stroke

Six trials involving 28,249 patients reported transient ischemic attack or stroke outcomes, which were combined in this pooled analysis [26–29,31,32]. Cumulative OR showed no differences between aspirin and no aspirin (0.6% vs. 0.6%; OR 0.96 with CI 0.71–1.30; I^2 = 0%, p = 0.81; Fig. 1D). A sensitivity analysis with exclusion of PEP for its low-risk CVD population resulted in similar results (Supplementary Fig. 3C).

3.3.3. Peripheral arterial events

Three trials reported peripheral arterial thrombotic events in 10,353 patients [26,27,29]. The meta-analysis showed no differences between aspirin and no aspirin groups (0.2% vs. 0.3%; OR 0.88 with CI 0.44–1.77; I^2 = 0%; p = 0.73; data not shown).

3.4. Venous thromboembolic events

Four trials comprising 27,797 patients reported data on venous thrombotic events including deep-vein thrombosis (DVT) and/or pulmonary embolism [26,27,29,32]. Reported details on additional anticoagulant medication are shown in Table 1. Venous

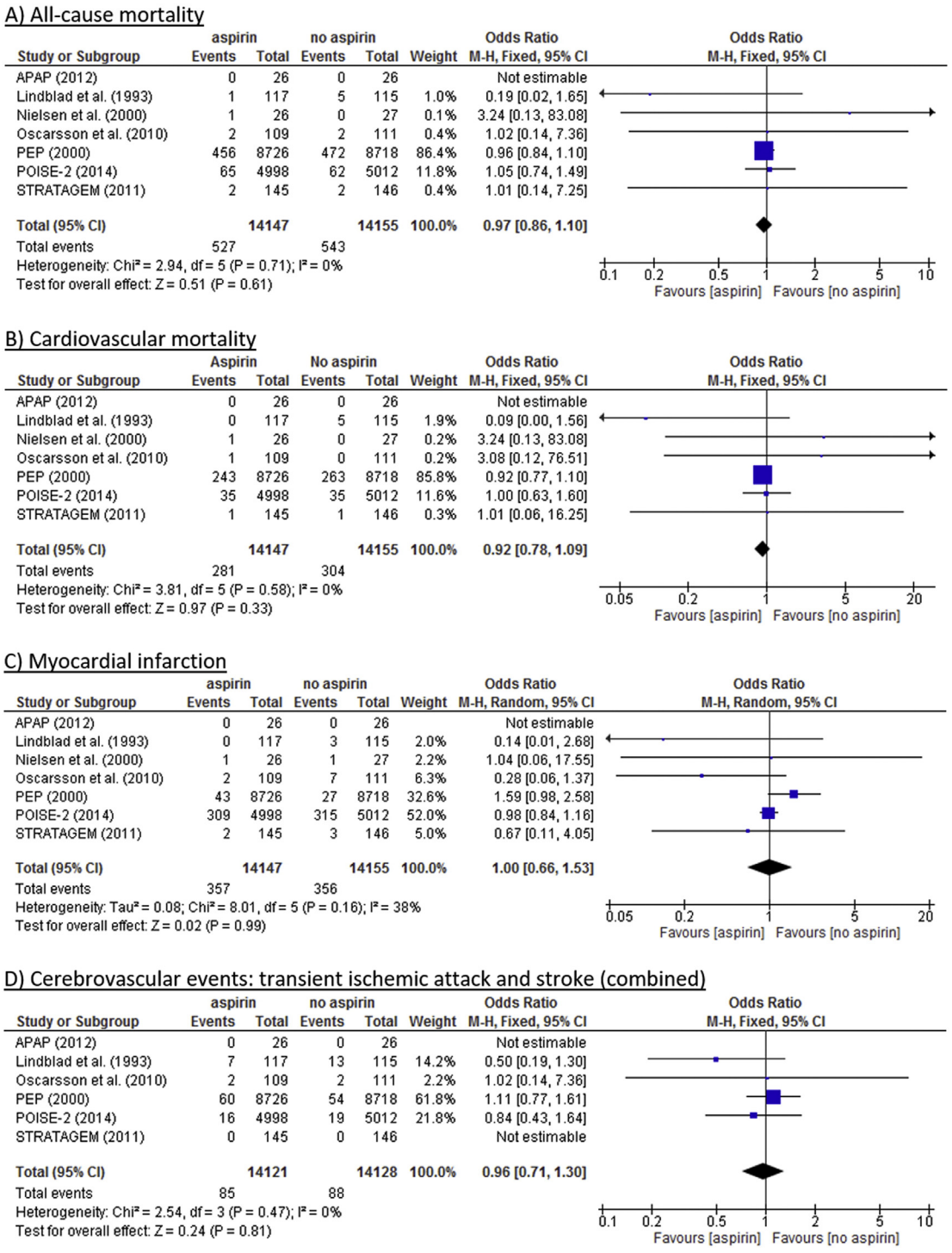


Fig. 1. Individual and summary odds ratios with 95% confidence intervals for all-cause mortality, cardiovascular mortality and arterial ischemic events in patients comparing aspirin vs. no aspirin. A) All-cause mortality; B) cardiovascular mortality; C) myocardial infarction; D) cerebrovascular events: transient ischemic attack and stroke (combined). M–H = Mantel–Haenszel; I^2 describes heterogeneity among studies, random-effects model used for $I^2 > 20\%$.

thrombotic events occurred significantly less in aspirin vs. no aspirin groups (0.8% vs. 1.1%; OR 0.74 with CI 0.58–0.94; $I^2 = 0\%$, $p = 0.02$; Fig. 2A). Cumulative OR for pulmonary embolism showed no differences (0.6% (aspirin) vs. 0.8% (no aspirin); OR 0.77 with CI 0.43–1.38; $I^2 = 74\%$; $p = 0.38$; Fig. 2B). The pooled analysis of combined thromboembolic events (venous thrombotic events + pulmonary embolism) resulted in a 25% relative risk reduction through aspirin (1.5% vs. 2.0%; OR 0.74 with CI 0.59–0.94; $I^2 = 21\%$; $p = 0.02$; Fig. 2C).

3.5. Major bleeding

Major bleeding was reported for all 28,302 patients in all seven trials [26–32] (Fig. 3). Cumulative episodes of major bleeding were increased

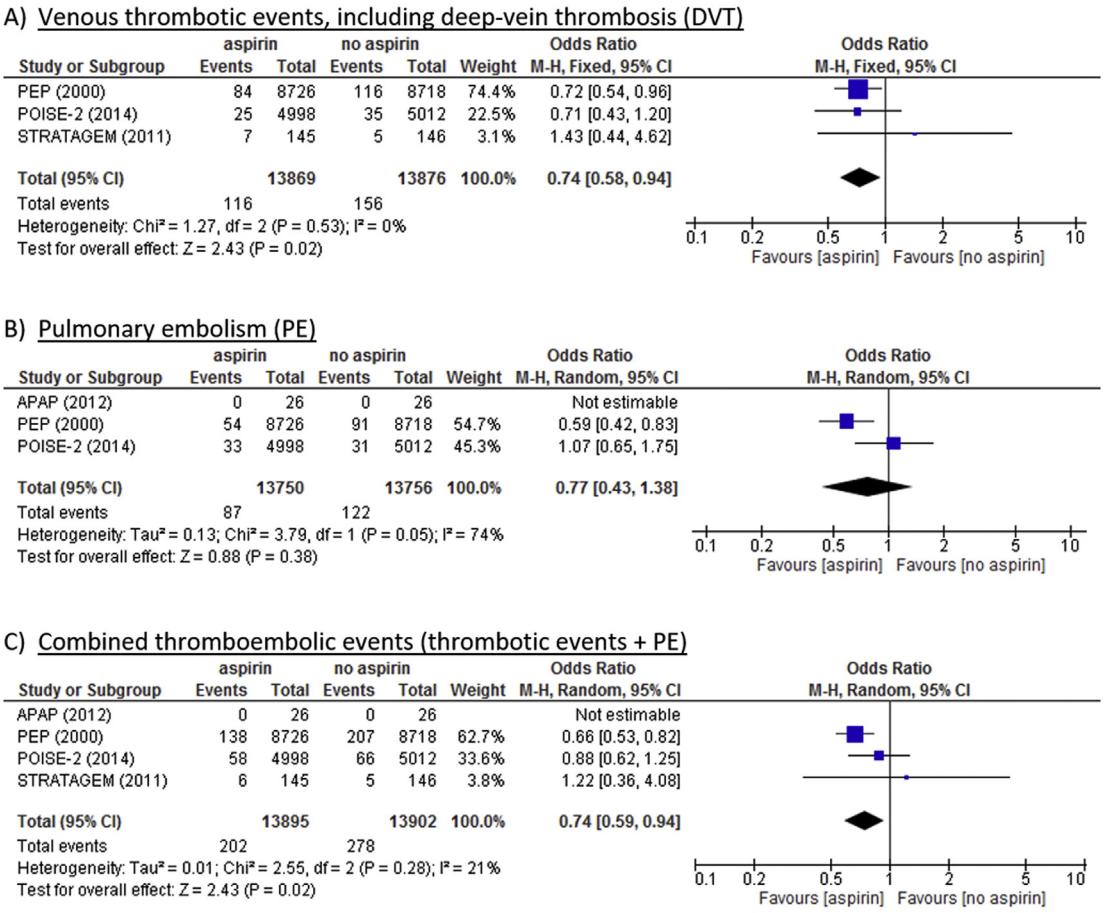


Fig. 2. Individual and summary odds ratios with 95% confidence intervals for venous thromboembolism. A) Venous thrombotic events, including deep-vein thrombosis (DVT); B) pulmonary embolism (PE); C) combined venous thrombotic events + PE. M–H = Mantel–Haenszel; I^2 describes heterogeneity among studies; random-effects model used for $I^2 > 20\%$.

by 16% in the aspirin group (4.3% vs. 3.7%; OR = 1.18 with CI 1.05–1.33; $I^2 = 0\%$; $p = 0.007$). A similar result could be obtained in a sensitivity analysis of CVD prevention trials with exclusion of PEP (6.4% vs. 5.3%; OR = 1.23 with CI 1.04–1.44; $I^2 = 0\%$; $p = 0.01$; Supplementary Fig. 3D). As the singular trial evaluating prevention of thromboembolism, PEP reported no significant increase in overall bleeding events under aspirin, although hip fracture patients alone had shown a significant increase in bleeding requiring transfusions in PEP [32].

3.6. Numbers needed to treat/to harm by perioperative aspirin therapy

Absolute risk reduction/increase and number needed to treat/to harm by treatment with aspirin for the significant outcomes of venous thromboembolism and major bleeding were calculated from event rates of the cumulative analyses (Fig. 2C and 3). 185 patients needed to be treated with aspirin to prevent a thromboembolic event, while 161 needed to be treated with aspirin to cause one additional major bleeding.

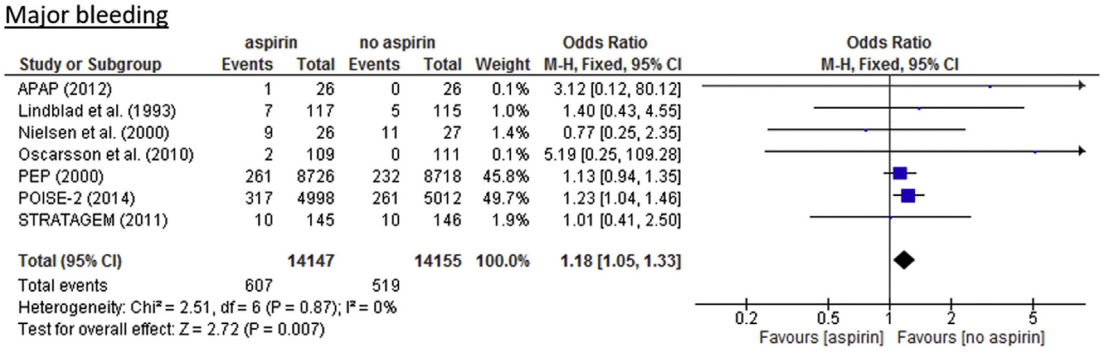


Fig. 3. Individual and summary odds ratios with 95% confidence intervals for perioperative major bleeding outcome. Fixed-effects model; M–H = Mantel–Haenszel; I^2 describes heterogeneity among studies.

4. Discussion

We here present a systematic review and meta-analysis of randomized controlled trials on efficacy and safety of perioperative aspirin vs. no aspirin therapy in non-cardiac surgery. The main findings are:

1) Aspirin therapy remained neutral with respect to perioperative all-cause and cardiovascular mortality; 2) aspirin showed no significant reduction in perioperative arterial ischemic event rates (myocardial infarction, cerebrovascular events, peripheral arterial events); 3) aspirin significantly reduced the odds of perioperative venous thromboembolic events; 4) aspirin significantly increased perioperative major bleedings.

4.1. Perioperative all-cause and cardiovascular mortality

Perioperative mortality reported in population-based registries with millions of non-cardiac surgical procedures amounts to 1–2% [3,33] with a tendency to decline [3,5]. Cardiovascular events are estimated to be responsible for about one third of all perioperative deaths [34–36]. Our analyzed cohorts showed an all-cause mortality rate of 3.7% and a cardiovascular mortality rate of 2.0%. This was above average but may be explained by the lack of low-risk surgery in the included trials. No singular trial found a significant influence of aspirin therapy on either outcome, which was also true for the pooled analysis: aspirin remained neutral both for all-cause mortality and cardiovascular mortality (Fig. 1). This was consistent in several performed sensitivity analyses (Supplementary Figs. 2 and 3A).

4.2. Aspirin for prevention of arterial ischemic events

The aspirin effect in arterial vessels is reflected in increased plaque stability, attenuation of vascular inflammation [37] and reduction of non-fatal coronary and cerebrovascular ischemic events in patients with known CVD [9–11,38]. Perioperative cardiovascular complications include myocardial infarction, stroke/transient ischemic attack and peripheral arterial ischemic events. CVD events have major impact on perioperative mortality risk [34,35,39,40]. Aspirin discontinuation before surgery was initially found to increase the risk for arterial ischemic events in observational studies and meta-analyses [41–44]. However – although underpowered for singular clinical outcomes – POISE-2 [27] showed no difference in the primary combined endpoint but instead an increased risk of major bleeding under aspirin. Including this POISE-2 finding, the European Society of Cardiology (ESC) [1] recommends considering aspirin discontinuation in the case of anticipated difficult hemostasis during surgery (Class IIa, level of evidence (LOE) B), however continuation may be considered based on an individual evaluation of bleeding risk against thrombotic complications (Class IIb, LOE B). The American College of Cardiology/American Heart Association (ACC/AHA) guidelines [45] state that initiation or continuation of aspirin is not beneficial in patients undergoing elective non-cardiac-non-carotid surgery (Class III, LOE C), who have not had previous coronary stenting, unless the risk of ischemic events outweighs the risk of surgical bleeding [45]. The Canadian Cardiovascular Society (CCS) guidelines take the most progressive approach, strongly recommending against initiation of aspirin before surgery for prevention of cardiac events, and furthermore recommending discontinuation three days before surgery except in patients with a recent coronary artery stent or undergoing carotid endarterectomy [46].

Myocardial infarction was the most frequent arterial ischemic event (2.5%) in our analysis, followed by cerebrovascular events (0.6%) and peripheral arterial events (0.2%). These event rates are below expected numbers [34,35], which may be related to individual study design: PEP did not routinely monitor postoperative troponin levels, resulting in a lower event rate [32,39].

The meta-analysis showed no significant differences between aspirin and no aspirin groups in the odds for myocardial infarction, cerebrovascular events and peripheral arterial events (Fig. 1). This was still true after elimination of the PEP trial from the sensitivity analyses because of its low cardiovascular risk population [32]. Lindblad et al. [28] solely tended towards benefits of aspirin in the setting of carotid endarterectomy. These findings are in line with conservative guideline recommendations in non-high-risk CVD patients, and especially support the 2017 CCS guideline recommendations [46].

4.3. Aspirin for prevention of venous thromboembolic events (VTE)

Aspirin-induced platelet inhibition attenuates venous thrombus formation: the PEP trial [32] was conducted in orthopedic surgery patients and revealed a significant reduction in the risk of deep-vein thrombosis (DVT) and pulmonary embolism (PE) under perioperative aspirin therapy. PEP results were reaffirmed in meta-analyses [47,48]. In non-orthopedic surgery, randomized trial data are lacking and alternative antithrombotic agents performed superior to aspirin [48]. The current American College of Chest Physicians (ACCP) 9th edition guidelines on perioperative VTE prophylaxis for orthopedic [49] surgery thus recommend aspirin for antithrombotic prophylaxis (Grade 1B) alongside low-molecular-weight heparin (LMWH), unfractionated heparin (UH), fondaparinux, apixaban etc. In non-orthopedic [50] surgery however, aspirin is only recommended in patients at high risk for VTE in whom both LMWH and UH are contraindicated (Grade 2C).

Our meta-analysis (Fig. 2) resulted in a significant 25% relative risk reduction of perioperative VTE under aspirin vs. no aspirin therapy, which was mainly driven by a reduction of venous thrombotic events, while PE showed no differences. Use of concomitant anticoagulants varied in studies (Table 1), which may explain differences between PEP and other trials (additional anticoagulants in PEP: 44.4%; POISE-2: 65%; STRATAGEM: 100%), possibly explaining the lower benefit in our analysis compared to PEP. The calculated NNT with aspirin was 185 for prevention of one VTE. These findings make aspirin a potential agent for VTE prophylaxis — at least in orthopedic surgery, as PEP and POISE-2 contribute 21,288 orthopedic patients to this analysis (75%). This is in line with contemporary large retrospective cohort analyses in orthopedic surgery [51].

4.4. Bleeding risk under perioperative aspirin therapy

Bleeding events are frequent perioperative complications with 10–30% [5,52] and are likewise associated with mortality [53–55]. Aspirin therapy increases spontaneous [10,11] and surgical [41] bleeding risk. Interestingly in POISE-2, a major bleeding event was an independent predictor of myocardial infarction [36]. Similar results have been obtained in PEP, where the aspirin group suffered more major bleedings and had an almost-significantly increased risk for MI (Figs. 1C and 3). Bleeding may potentially cause cardiac events by circulatory impairment and increased cardiac workload through loss of blood and oxygen supply capacity [36,56], and may thus be a cardiovascular risk factor in itself.

The pooled analysis of all trials showed a significant 17% increase in the risk of major bleeding under perioperative aspirin vs. no aspirin therapy (Fig. 3). Patients at risk for CVD (6 trials) showed a 21% bleeding risk increase under aspirin. Aspirin's calculated NNH was 161 for causing one major bleeding. Only Nielsen et al. [30] found

less bleeding events under aspirin — which may be related to the high-risk-of-bleeding setting of transurethral prostatectomy and the low patient-count [57]. Our analysis is in line with a meta-analysis of PEP and POISE-2 performed by the POISE-2 authors [47], and low statistical heterogeneity further confirms validity of results.

4.5. Benefit/risk evaluation of perioperative aspirin therapy

A patient-tailored recommendation for perioperative aspirin management has to take aspirin's effect on arterial ischemia, on venous thromboembolism and on bleeding in the perioperative setting into account, and weigh it carefully against the individual risk profile of the respective patient.

Survival is the most important and best-measured endpoint and integrates all treatment effects: in an all-comer scenario of intermediate-risk CVD patients, aspirin therapy remains neutral to survival and cardiovascular mortality. It also holds no significant benefit for prevention of arterial ischemic events. A beneficial effect on VTE in (orthopedic) surgical patients may justify this indication, although the risk for major bleedings is at least evenly increased.

The present analysis thus suggests avoiding aspirin continuation/initiation in non-cardiac surgery in the average CVD patient. Aspirin may have its perioperative benefits for cardiovascular high-risk patients with previous MI, coronary stenting or bypass surgery (CABG), as a recently presented post-hoc analysis of POISE-2 patients with prior PCI suggests [58]. However, those patients are not sufficiently represented in available RCTs yet: adequate risk classification for their identification would allow a more individualized approach and should be a specific focus of future RCTs in high-risk CVD patients. These could potentially profit from perioperative aspirin therapy, as the balance of thromboembolic/ischemic benefits to bleeding risks may here have an overall effect on survival.

4.6. Limitations

All meta-analyses are characterized by heavy statistical weight on PEP [32] and POISE-2 [27] trials, which contributed 91% of patients to the cumulative patient number. Both were high-quality studies, published in high-class journals and with a low risk of bias. However, due to their dissimilar protocols and patient populations, direct comparisons are difficult. Additionally, all included studies are heterogenous concerning the range of surgical procedures and bleeding risk. While this considerably reduces accuracy of calculations for a specific setting, it reflects a very large real-life surgical population and improves generalizability. We accounted for these natural limitations with sensitivity analyses and careful interpretation of results.

Availability of patient-level instead of study-level data would have considerably improved accuracy of results and allowed subgroup comparisons. This limitation is also reflected in incomplete reports of study patient characteristics and endpoint definitions (Supplementary Tables 1 and 3), as well as details of concomitant anticoagulation and/or P2Y12-platelet-inhibitors.

5. Conclusion

This meta-analysis of randomized controlled trials of aspirin vs. no aspirin therapy in non-cardiac surgery found aspirin to remain neutral with respect to overall survival as well as cardiovascular mortality. Arterial ischemic events showed no differences, while venous thromboembolic events were significantly reduced by aspirin therapy, albeit at the expense of perioperative major bleedings. Thus, the available data support recommendations against perioperative aspirin continuation/initiation in the average CVD patient at intermediate cardiovascular risk, as well as recommendations of aspirin for perioperative VTE prophylaxis in orthopedic patients only. Individualized therapies for cardiovascular high-risk patients potentially profiting from perioperative aspirin need to be established.

Contributors

VS, EPN and GW conceived and designed the study. MB, YL, GW, MiK and VS collected sources, selected studies and abstracted data. GW and EPN performed the statistical analyses. GW, VS, FA and MK drafted the manuscript. All authors analyzed and interpreted the data and critically revised the manuscript. All authors read and accepted the submitted version of the manuscript.

Competing interests

All authors have completed the ICMJE uniform disclosure form at www.icmje.org/coi_disclosure.pdf and declare: no support from any organization for the submitted work; no financial relationships with any organizations that might have an interest in the submitted work in the previous three years; and no other relationships or activities that could appear to have influenced the submitted work.

Registration

Not registered.

Funding source

None of the authors received third-party funding for this work. Design, conduct and reporting are thus free from financial influence.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijcard.2017.12.088.

References

[1] S.D. Kristensen, J. Knuuti, A. Saraste, S. Anker, H.E. Botker, S. De Hert, et al., ESC/ESA guidelines on non-cardiac surgery: cardiovascular assessment and management: the joint task force on non-cardiac surgery: cardiovascular assessment and management of the European Society of Cardiology (ESC) and the European Society of Anaesthesiology (ESA), Eur. Heart J. 2014 (2014).
[2] T.G. Weiser, S.E. Regenbogen, K.D. Thompson, A.B. Haynes, S.R. Lipsitz, W.R. Berry, et al., An estimation of the global volume of surgery: a modelling strategy based on available data, Lancet 372 (2008) 139–144.
[3] M.E. Semel, S.R. Lipsitz, L.M. Funk, A.M. Bader, T.G. Weiser, A.A. Gawande, Rates and patterns of death after surgery in the United States, 1996 and 2006, Surgery 151 (2012) 171–182.
[4] A.B. Haynes, T.G. Weiser, W.R. Berry, S.R. Lipsitz, A.H. Breizat, E.P. Dellinger, et al., A surgical safety checklist to reduce morbidity and mortality in a global population, N. Engl. J. Med. 360 (2009) 491–499.
[5] A.A. Gawande, E.J. Thomas, M.J. Zinner, T.A. Brennan, The incidence and nature of surgical adverse events in Colorado and Utah in 1992, Surgery 126 (1999) 66–75.
[6] M.A. Healey, S.R. Shackford, T.M. Osler, F.B. Rogers, E. Burns, Complications in surgical patients, Arch. Surg. 137 (2002) 611–617 (discussion 7-8).

[7] H. Levesque, O. Lafont, Aspirin throughout the ages: a historical review, Rev. Med. Interne. 21 (Suppl. 1) (2000) 8s–17s.
[8] J.R. Vane, Inhibition of prostaglandin synthesis as a mechanism of action for aspirin-like drugs, Nat. New. Biol. 231 (1971) 232–235.
[9] Collaborative overview of randomised trials of antiplatelet therapy—II. Maintenance of vascular graft or arterial patency by antiplatelet therapy. Antiplatelet Trialists' Collaboration, BMJ 308 (1994) 159–168.
[10] C. Antithrombotic Trialists, Collaborative meta-analysis of randomised trials of antiplatelet therapy for prevention of death, myocardial infarction, and stroke in high risk patients, BMJ 324 (2002) 71–86.
[11] C. Antithrombotic Trialists, C. Baigent, L. Blackwell, R. Collins, J. Emberson, J. Godwin, et al., Aspirin in the primary and secondary prevention of vascular disease: collaborative meta-analysis of individual participant data from randomised trials, Lancet 373 (2009) 1849–1860.
[12] M.F. Piepoli, A.W. Hoes, S. Agewall, C. Albus, C. Brotons, A.L. Catapano, et al., 2016 European guidelines on cardiovascular disease prevention in clinical practice: the Sixth Joint Task Force of the European Society of Cardiology and Other Societies on Cardiovascular Disease Prevention in Clinical Practice (constituted by representatives of 10 societies and by invited experts). Developed with the special contribution of the European Association for Cardiovascular Prevention & Rehabilitation (EACPR), Eur. Heart J. 37 (2016) 2315–2381.
[13] S.C. Smith Jr., E.J. Benjamin, R.O. Bonow, L.T. Braun, M.A. Creager, B.A. Franklin, et al., AHA/ACCF secondary prevention and risk reduction therapy for patients with coronary and other atherosclerotic vascular disease: 2011 update: a guideline from the American Heart Association and American College of Cardiology Foundation, Circulation 124 (2011) 2458–2473.
[14] Collaborative overview of randomised trials of antiplatelet therapy—I. Prevention of death, myocardial infarction, and stroke by prolonged antiplatelet therapy in various categories of patients. Antiplatelet Trialists' Collaboration, BMJ 308 (1994) 81–106.
[15] G. De Berardis, G. Lucisano, A. D'Ettorre, F. Pellegrini, V. Lepore, G. Tognoni, et al., Association of aspirin use with major bleeding in patients with and without diabetes, JAMA 307 (2012) 2286–2294.
[16] C. Becattini, G. Agnelli, A. Schenone, S. Eichinger, E. Bucherini, M. Silingardi, et al., Aspirin for preventing the recurrence of venous thromboembolism, N. Engl. J. Med. 366 (2012) 1959–1967.
[17] T.A. Brighton, J.W. Eikelboom, K. Mann, R. Mister, A. Gallus, P. Ockelford, et al., Low-dose aspirin for preventing recurrent venous thromboembolism, N. Engl. J. Med. 367 (2012) 1979–1987.
[18] S.V. Konstantinides, A. Torbicki, G. Agnelli, N. Danchin, D. Fitzmaurice, N. Galie, et al., ESC guidelines on the diagnosis and management of acute pulmonary embolism, Eur. Heart J. 35 (2014) 3033–3069 (69a-69k).
[19] Collaborative overview of randomised trials of antiplatelet therapy—III. Reduction in venous thrombosis and pulmonary embolism by antiplatelet prophylaxis among surgical and medical patients. Antiplatelet Trialists' Collaboration, BMJ 308 (1994) 235–246.
[20] A. Liberati, D.G. Altman, J. Tetzlaff, C. Mulrow, P.C. Gotzsche, J.P. Ioannidis, et al., The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration, BMJ 339 (2009) b2700.
[21] J.P. Higgins, S. Green, Cochrane handbook for systematic reviews of interventions, version 5.1.0 (updated March 2011). The Cochrane Collaboration, 2011 Available from www.handbook.cochrane.org 2011.
[22] R. Mehran, S.V. Rao, D.L. Bhatt, C.M. Gibson, A. Caixeta, J. Eikelboom, et al., Standardized bleeding definitions for cardiovascular clinical trials: a consensus report from the Bleeding Academic Research Consortium, Circulation 123 (2011) 2736–2747.
[23] J.P. Higgins, S.G. Thompson, J.J. Deeks, D.G. Altman, Measuring inconsistency in meta-analyses, BMJ 327 (2003) 557–560.
[24] J.E. Grizzle, Analysis of data from multiclinic trials, Control. Clin. Trials 8 (1987) 392–393.
[25] R. DerSimonian, N. Laird, Meta-analysis in clinical trials revisited, Contemp. Clin. Trials 45 (2015) 139–145.
[26] D. Antolovic, A. Rakow, P. Contin, A. Ulrich, N.N. Rahbari, M.W. Buchler, et al., A randomised controlled pilot trial to evaluate and optimize the use of anti-platelet agents in the perioperative management in patients undergoing general and abdominal surgery—the APAP trial (ISRCTN45810007), Langenbeck's Arch. Surg. 397 (2012) 297–306.
[27] P.J. Devereaux, M. Mrkobrada, D.I. Sessler, K. Leslie, P. Alonso-Coello, A. Kurz, et al., Aspirin in patients undergoing noncardiac surgery, N. Engl. J. Med. 370 (2014) 1494–1503.
[28] B. Lindblad, N.H. Persson, R. Takolander, D. Bergqvist, Does low-dose acetylsalicylic acid prevent stroke after carotid surgery? A double-blind, placebo-controlled randomized trial, Stroke 24 (1993) 1125–1128.
[29] J. Mantz, C.M. Samama, F. Tubach, P.J. Devereaux, J.P. Collet, P. Albaladejo, et al., Impact of preoperative maintenance or interruption of aspirin on thrombotic and bleeding events after elective non-cardiac surgery: the multicentre, randomized, blinded, placebo-controlled, STRATAGEM trial, Br. J. Anaesth. 107 (2011) 899–910.
[30] J.D. Nielsen, A. Holm-Nielsen, J. Jespersen, C.C. Vinther, I.W. Settgast, J. Gram, The effect of low-dose acetylsalicylic acid on bleeding after transurethral prostatectomy—a prospective, randomized, double-blind, placebo-controlled study, Scand. J. Urol. Nephrol. 34 (2000) 194–198.
[31] A. Oscarsson, A. Gupta, M. Fredrikson, J. Jarhult, M. Nystrom, E. Pettersson, et al., To continue or discontinue aspirin in the perioperative period: a randomized, controlled clinical trial, Br. J. Anaesth. 104 (2010) 305–312.
[32] Prevention of pulmonary embolism and deep vein thrombosis with low dose aspirin: Pulmonary Embolism Prevention (PEP) trial, Lancet 355 (2000) 1295–1302.
[33] P.G. Noordzij, D. Poldermans, O. Schouten, J.J. Bax, F.A. Schreiner, E. Boersma, Postoperative mortality in The Netherlands: a population-based analysis of surgery-specific risk in adults, Anesthesiology 112 (2010) 1105–1115.
[34] F. Botto, P. Alonso-Coello, M.T. Chan, J.C. Villar, D. Xavier, S. Srinathan, et al., Myocardial injury after noncardiac surgery: a large, international, prospective cohort study establishing diagnostic criteria, characteristics, predictors, and 30-day outcomes, Anesthesiology 120 (2014) 564–578.
[35] P.J. Devereaux, M.T. Chan, P. Alonso-Coello, M. Walsh, O. Berwanger, J.C. Villar, et al., Association between postoperative troponin levels and 30-day mortality among patients undergoing noncardiac surgery, JAMA 307 (2012) 2295–2304.
[36] P.J. Devereaux, D.I. Sessler, Cardiac complications in patients undergoing major noncardiac surgery, N. Engl. J. Med. 373 (2015) 2258–2269.
[37] T. Cyrus, S. Sung, L. Zhao, C.D. Funk, S. Tang, D. Pratico, Effect of low-dose aspirin on vascular inflammation, plaque stability, and atherogenesis in low-density lipoprotein receptor-deficient mice, Circulation 106 (2002) 1282–1287.
[38] Framework MRCsGPR, Thrombosis prevention trial: randomised trial of low-intensity oral anticoagulation with warfarin and low-dose aspirin in the primary prevention of ischaemic heart disease in men at increased risk. The Medical Research Council's General Practice Research Framework, Lancet 351 (1998) 233–241.
[39] P.J. Devereaux, D. Xavier, J. Pogue, G. Guyatt, A. Sigamani, I. Garutti, et al., Characteristics and short-term prognosis of perioperative myocardial infarction in patients undergoing noncardiac surgery: a cohort study, Ann. Intern. Med. 154 (2011) 523–528.
[40] P.J. Devereaux, Relationship between high sensitive troponin T measurements and 30-day mortality after noncardiac surgery, Presented at ACC 2017, 2017 (March 19, 2017).
[41] W. Burger, J.M. Chemnitius, G.D. Kneissl, G. Rucker, Low-dose aspirin for secondary cardiovascular prevention — cardiovascular risks after its perioperative withdrawal versus bleeding risks with its continuation — review and meta-analysis, J. Intern. Med. 257 (2005) 399–414.
[42] J.P. Collet, F. Himbet, P.G. Steg, Myocardial infarction after aspirin cessation in stable coronary artery disease patients, Int. J. Cardiol. 76 (2000) 257–258.
[43] E. Ferrari, M. Benhamou, P. Cerboni, B. Marcel, Coronary syndromes following aspirin withdrawal: a special risk for late stent thrombosis, J. Am. Coll. Cardiol. 45 (2005) 456–459.
[44] K. Senior, Aspirin withdrawal increases risk of heart problems, Lancet 362 (2003) 1558.
[45] L.A. Fleisher, K.E. Fleischmann, A.D. Auerbach, S.A. Barnason, J.A. Beckman, B. Bozkurt, et al., ACC/AHA guideline on perioperative cardiovascular evaluation and management of patients undergoing noncardiac surgery: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines, J. Am. Coll. Cardiol. 2014 (2014).
[46] E. Duceppe, J. Parlow, P. MacDonald, K. Lyons, M. McMullen, S. Srinathan, et al., Canadian Cardiovascular Society guidelines on perioperative cardiac risk assessment and management for patients who undergo noncardiac surgery, Can. J. Cardiol. 33 (2017) 17–32.
[47] J.W. Eikelboom, C. Kearon, G. Guyatt, D.I. Sessler, S. Yusuf, D. Cook, et al., Perioperative aspirin for prevention of venous thromboembolism: the PeriOperative ISchemia Evaluation-2 trial and a pooled analysis of the randomized trials, Anesthesiology 125 (2016) 1121–1129.
[48] S.M. Sahebally, D. Healy, S.R. Walsh, Aspirin in the primary prophylaxis of venous thromboembolism in surgical patients, Surgeon 13 (2015) 348–358.
[49] Y. Falck-Ytter, C.W. Francis, N.A. Johanson, C. Curley, O.E. Dahl, S. Schulman, et al., Prevention of VTE in orthopedic surgery patients: antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines, Chest 141 (2012) e278S–325S.
[50] M.K. Gould, D.A. Garcia, S.M. Wren, P.J. Karanicolas, J.I. Arcelus, J.A. Heit, et al., Prevention of VTE in nonorthopedic surgical patients: antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines, Chest 141 (2012) (e227S–77S).
[51] J.N. Chu, J. Maselli, A.D. Auerbach, M.C. Fang, The risk of venous thromboembolism with aspirin compared to anticoagulants after hip and knee arthroplasty, Thromb. Res. 155 (2017) 65–71.
[52] M.E. Stokes, X. Ye, M. Shah, K. Mercaldi, M.W. Reynolds, M.F. Rupnow, et al., Impact of bleeding-related complications and/or blood product transfusions on hospital costs in inpatient surgical patients, BMC Health Serv. Res. 11 (2011) 135.
[53] J.L. Carson, R.M. Poses, R.K. Spence, G. Bonavita, Severity of anaemia and operative mortality and morbidity, Lancet 1 (1988) 727–729.
[54] R.K. Spence, J.A. Carson, R. Poses, S. McCoy, M. Pello, J. Alexander, et al., Elective surgery without transfusion: influence of preoperative hemoglobin level and blood loss on mortality, Am. J. Surg. 159 (1990) 320–324.
[55] Wu WC, T.S. Smith, W.G. Henderson, C.B. Eaton, R.M. Poses, G. Uttley, et al., Operative blood loss, blood transfusion, and 30-day mortality in older patients after major noncardiac surgery, Ann. Surg. 252 (2010) 11–17.
[56] L.G. Glance, A.W. Dick, D.B. Mukamel, F.J. Fleming, R.A. Zollo, R. Wissler, et al., Association between intraoperative blood transfusion and mortality and

morbidity in patients undergoing noncardiac surgery, Anesthesiology 114 (2011) 283–292.
[57] A.V. Thurston, S.L. Briant, Aspirin and post-prostatectomy haemorrhage, Br. J. Urol. 71 (1993) 574–576.
[58] M Graham, Aspirin in patients with previous percutaneous coronary intervention (PCI) undergoing noncardiac surgery: the POISE-2 PCI substudy, Presented at: American Heart Association 2017 Scientific Sessions, November 14, 2017 (Anaheim, CA).
[59] D. Antolovic, C. Reissfelder, A. Rakow, P. Contin, N.N. Rahbari, M.W. Buchler, et al., A randomised controlled trial to evaluate and optimize the use of antiplatelet agents in the perioperative management in patients undergoing general and abdominal surgery—the APAP trial (ISRCTN45810007), BMC Surg. 11 (2011) 7.